

Study of Entropy in Unitary Random Quantum Circuits with 2-Qubit Gates Architecture

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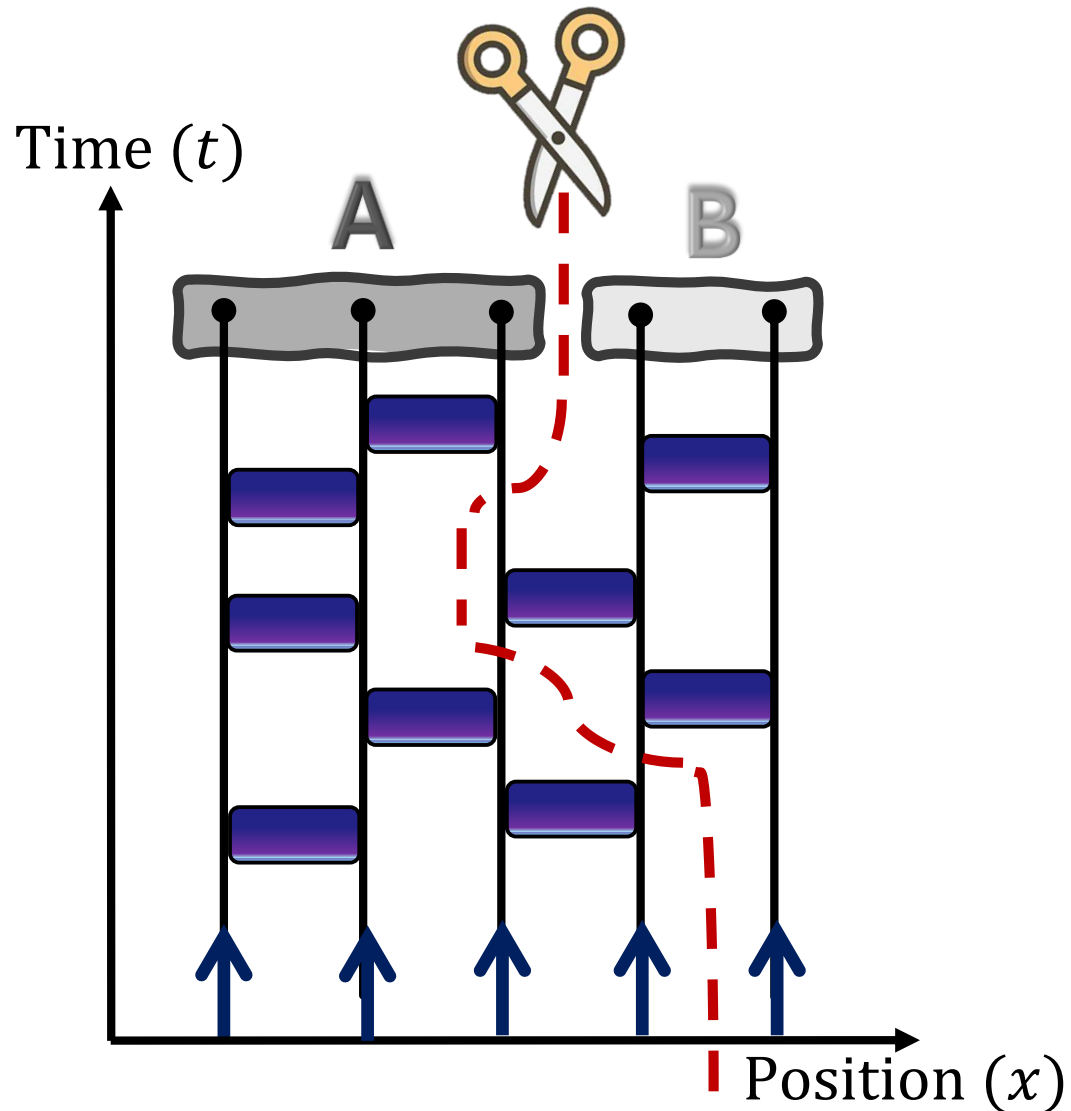


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MOTIVATION



Characterizing how entanglement grows with time in a many-body system, is a key problem in nonequilibrium quantum physics.

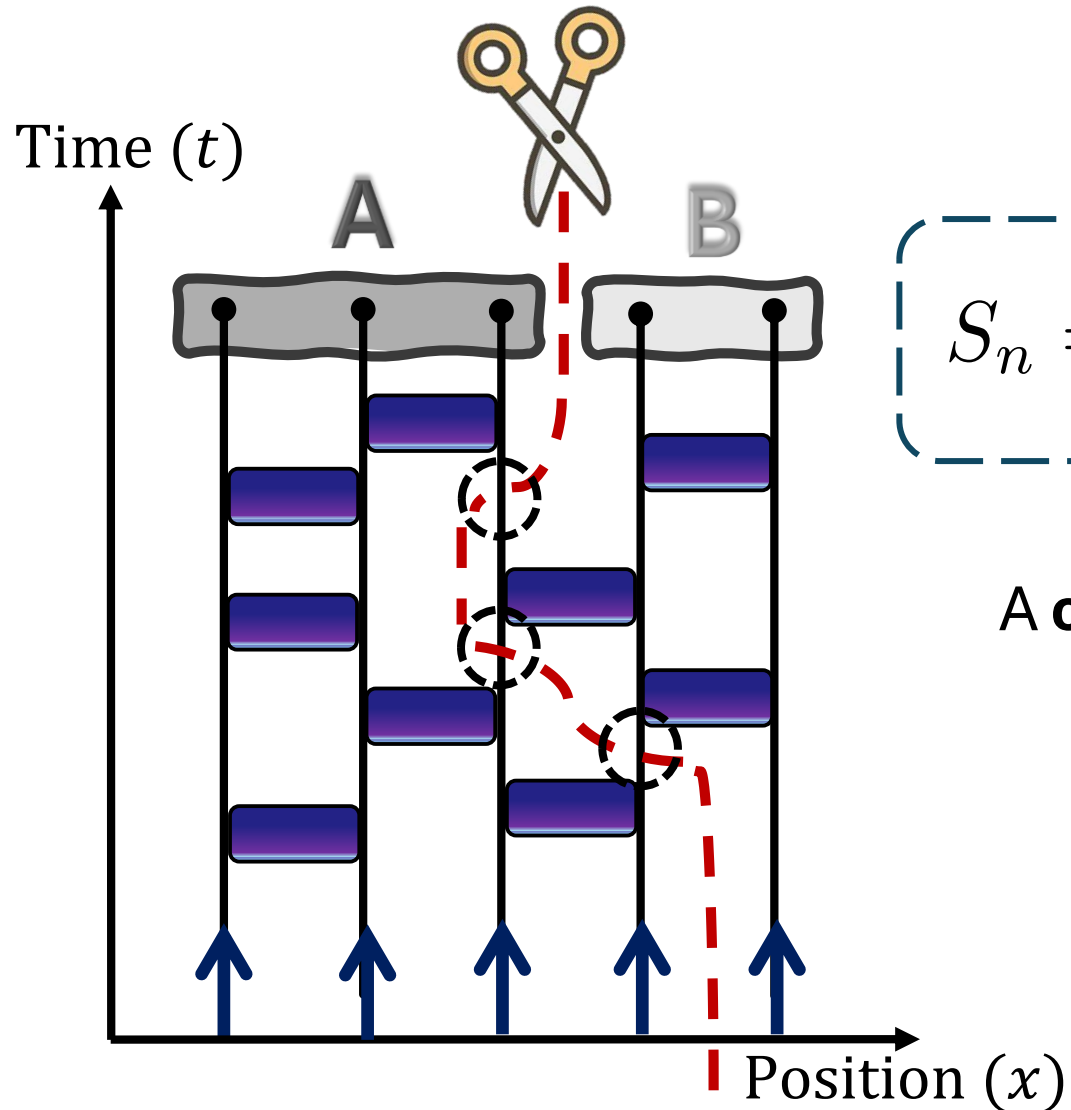


The minimal cut-formalism provides a classical bound to the entanglement growth in **1D-circuits**.

Fisher, M. P., Nahum, A., et al. (2023). Random quantum circuits. Annual Review of Condensed Matter Physics, 14(1), 335-379

Horodecki, Ryszard, et al. Quantum entanglement. Reviews of modern physics, 2009, vol. 81, no 2, p. 865-942.

MINIMAL CUT



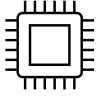
Rényi entropy

$$S_n = \frac{1}{1-n} \log (\text{Tr}(\rho_A^n)) = \frac{1}{1-n} \log \left(\sum_i \lambda_i^n \right)$$

A **classical** upper bound for the bipartite entropies,

$$S_n \leq N_{\text{min cut}}$$

TWO-QUBIT GATES CIRCUIT



When restricted to two-qubit (local) gates, and in the thermodynamic limit, the problem of minimizing the number of cuts is the same as that of finding the energy minima of a polymer in a disordered medium (DPRE).



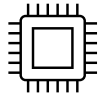
The entanglement entropy across a cut to grow according to the Kardar-Parisi-Zhang (KPZ) equation.

$$\frac{\partial S}{\partial t} = \nu \partial_x^2 S - \frac{\lambda}{2} (\partial_x S)^2 + \eta(x, t) + c$$

diffusive smoothing

noise

KPZ MODEL



KPZ scaling is characterized by an exponent β governing the size of fluctuations in the interface height, an exponent α governing spatial correlations.

- The mean entropy (entanglement) grows linearly in time, with a universal subleading correction:

$$\langle S(x, t) \rangle = v_E t + B t^\beta$$

- The fluctuations grow as:

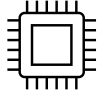
$$\langle \langle S(x, t)^2 \rangle \rangle^{1/2} = C t^\beta$$

In 1+1 dimension
 $\beta = 1/3, \alpha = 1/2$

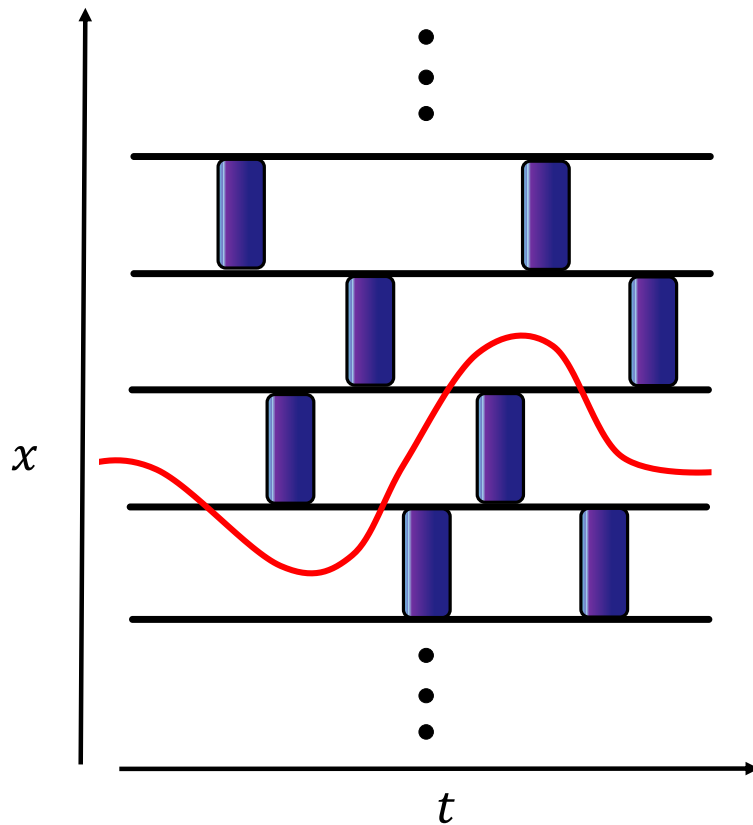
- The correlation length governing spatial correlations in the fluctuations grows with time as:

$$\xi(t) \sim t^{\beta/\alpha}$$

CUT GROWTH MODEL



The minimal cut, can be view as the word-line traced by a classical point-particle (walker) which avoids the obstacles.



the probability $p(x, t)$ that the “walker” is in the x -th carrel at time t ,

$$\begin{aligned} p(x, t + 1) = & \frac{1}{2} c(x + 1, t) p(x + 1, t) \\ & + \frac{1}{2} c(x - 1, t) p(x - 1, t) \\ & + (1 - c(x, t)) p(x, t) \end{aligned}$$

Where $c(x, t)$ is the probability that a two-qubit gate is in the x -th carrel, at time t .

CUT GROWTH MODEL

- We are interested in the cut increase distribution $Q(t)$, that is, the number of times the walker changes carrel by avoiding an obstacle:

$$Q(t) = \sum_x c(x, t) p(x, t)$$

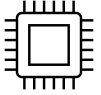
- And we also define $\mathcal{N}_c(t)$ as the cut number distribution in time:

$$\mathcal{N}_c(t) = \sum_{l=0}^t \Delta \mathcal{N}_c(l), \quad Q(t) = \text{Prob}(\Delta \mathcal{N}_c(t) = 1)$$

- Each cut-gap is an independent-binary random variable with distribution:

$$\varrho(\Delta \mathcal{N}_c(t)) = Q(t) \delta(\Delta \mathcal{N}_c(t) - 1) + (1 - Q(t)) \delta(\Delta \mathcal{N}_c(t))$$

CUT GROWTH MODEL



Therefore, the whole distribution P of the $\mathcal{N}_c(t)$ is given by:

$$P(\mathcal{N}_c(t)) \propto \int_{\mathbb{R}} e^{ik\mathcal{N}_c(t) + \sum_l^t \log(1 + Q(l)[e^{-ik} - 1])} dk$$

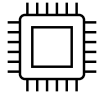
After integrating:

$$P(\mathcal{N}_c(t)) \propto \frac{1}{\mathcal{N}_c(t)!} \exp \left(\sum_{\tau=1}^t \log(1 - Q(\tau)) \right) \left(\sum_{\tau=1}^t \frac{Q(\tau)}{1 - Q(\tau)} \right)^{\mathcal{N}_c(t)}$$

For large enough number of steps, the number of cuts $\mathcal{N}_c(t)$ will increase as well:

$$P(\mathcal{N}_c(t)) \propto \exp \left(-\frac{1}{2t} \frac{(\mathcal{N}_c(t) - t\overline{Q}(t))^2}{(\overline{Q}(t) - \overline{Q^2}(t))} \right), \quad \overline{Q^m}(t) = \frac{1}{t} \sum_{\tau=1}^t Q^m(\tau)$$

CUT GROWTH MODEL



For the particular case of uniform distributed two-qubit gates, $c(x, t) = \frac{1}{L-1}$, with L the number of qubits,

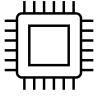
$$Q(t) = \frac{1}{L-1} \sum_x p(x, t)$$

Assuming large L , a Gaussian distribution is obtained for P , where the mean is given by $t \overline{Q}(t)$, therefore:

$$\langle \mathcal{N}_c(t) \rangle = \frac{t}{L}$$

However, more than the average cut, we are interested in the average minimal cut.

CUT GROWTH MODEL



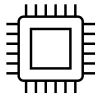
The minimal cut maintains the same behavior as the average cut and must be bounded by it;

$$\langle \mathcal{N}_{\text{minimal cut}}(t) \rangle \sim \frac{t}{L}$$

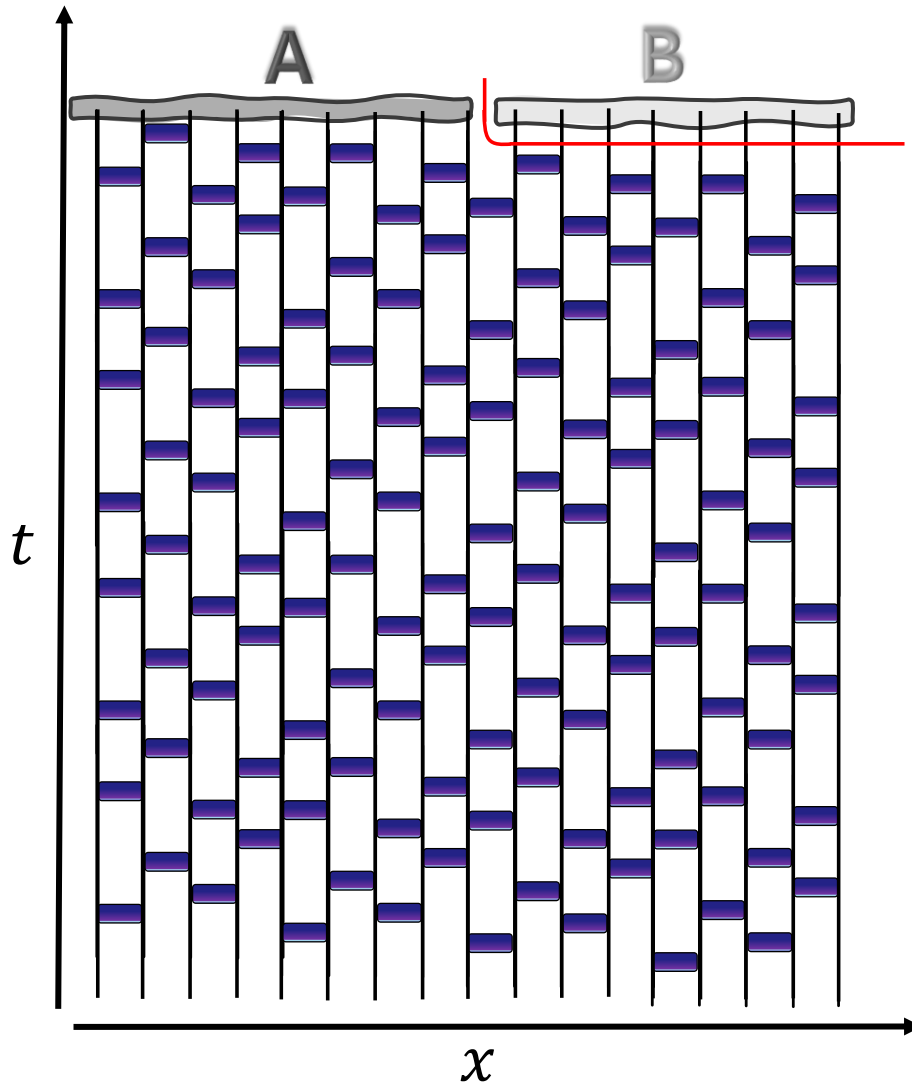
Where the entanglement velocity is to be determined as;

$$v_E = \frac{\nu}{L}, \nu \leq 1$$

SCALING FORM FOR ENTANGLEMENT SATURATION



Minimal cut is essentially vertical (prior to saturation of the entropy)

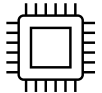


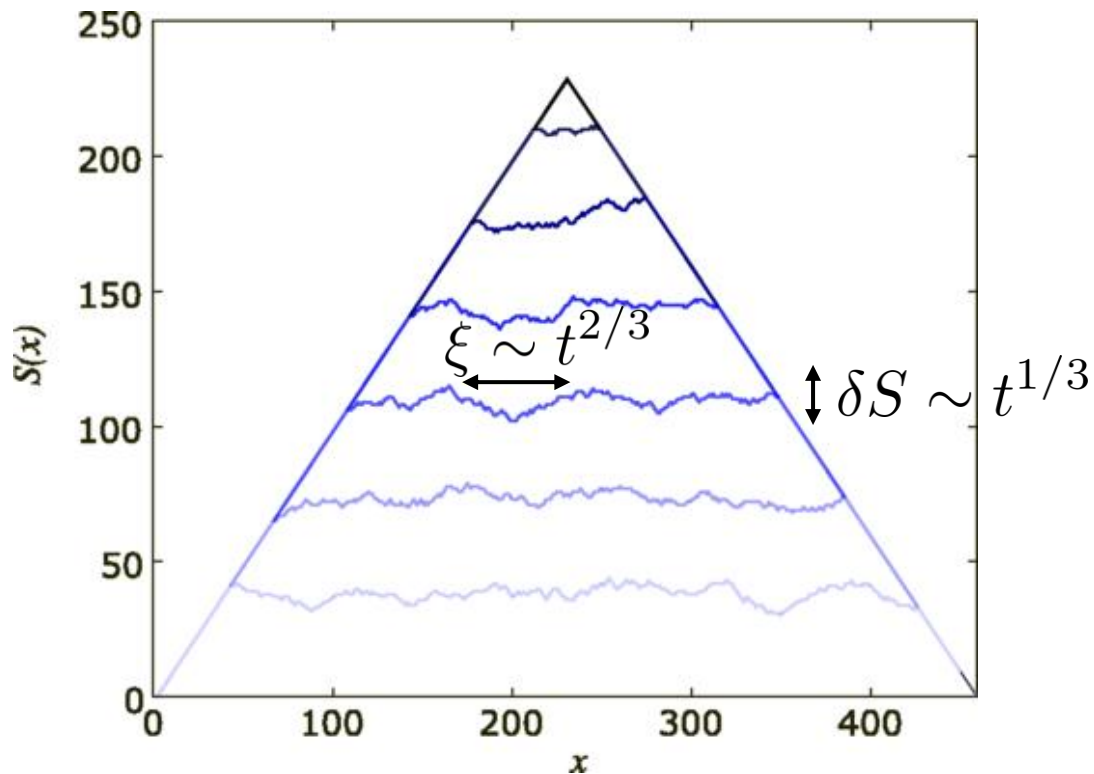
$$0 \leq S_n \leq \ln D, \text{ where } D = \min(D_A, D_B).$$

$$\hookrightarrow \boxed{0 \leq S_n(x, t) \leq x}$$

At short times, the entanglement growth is affected by initial conditions, while on very long-time scales, of the order of the system size, the entanglement saturates.

SCALING FORM FOR ENTANGLEMENT SATURATION

 In a finite system the asymptotic $\langle S(x, t) \rangle$ profile is that of a pyramid, with a maximum at height $x = L/2$, whose height is $L/2$.



The asymptotic behavior of entropy is governed by the minimal cut;

$$\langle S(x, t) \rangle \sim \langle \mathcal{N}_{\text{minimal cut}}(t) \rangle$$



$$\langle S(x, t) \rangle \sim \begin{cases} v_e t, & \text{for } x \leq t_{\text{saturation}} \\ x, & \text{for } x > t_{\text{saturation}} \end{cases}$$

$$t_{\text{saturation}} = \frac{x}{v_E} \approx \frac{L}{2v_E}$$

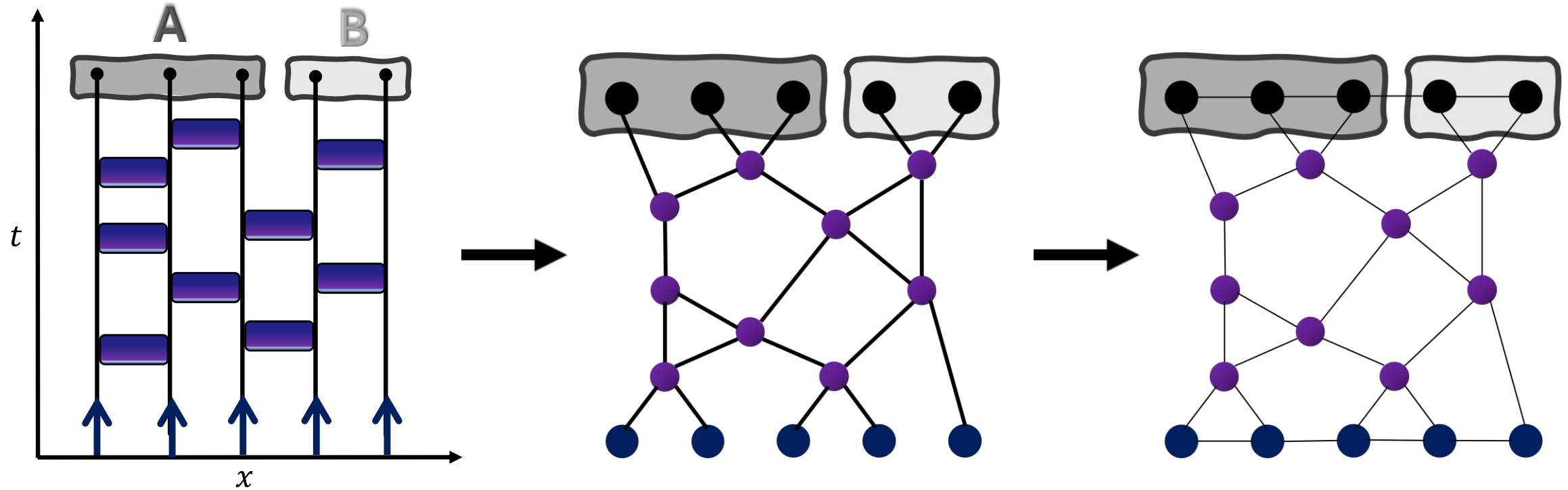
Numerical simulations of the entanglement entropy $S_1(x, t)$ for a random Clifford circuit with 459 qubits. Adapted from Ref. [1].

[1] Nahum, A., Ruhman, J., Vijay, S., & Haah, J. (2017). Quantum entanglement growth under random unitary dynamics. *Physical Review X* 7 (3), 031016.

SIMULATION

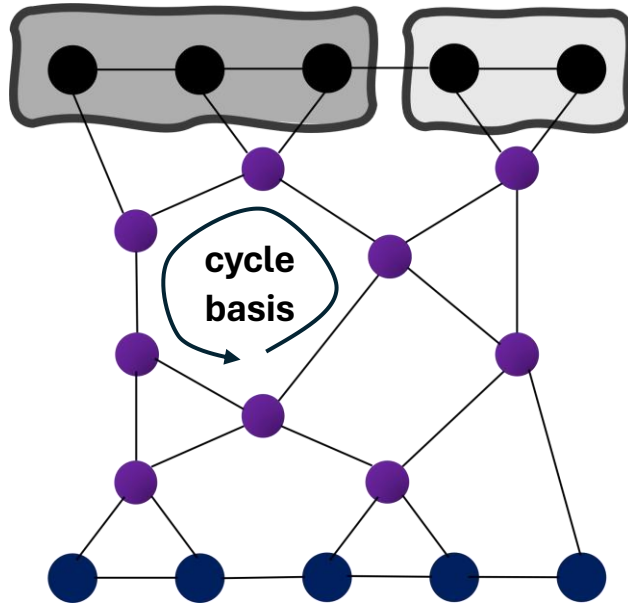
Algorithm for minimal cut

1. The problem of finding the minimal cut is mapped to a graph.



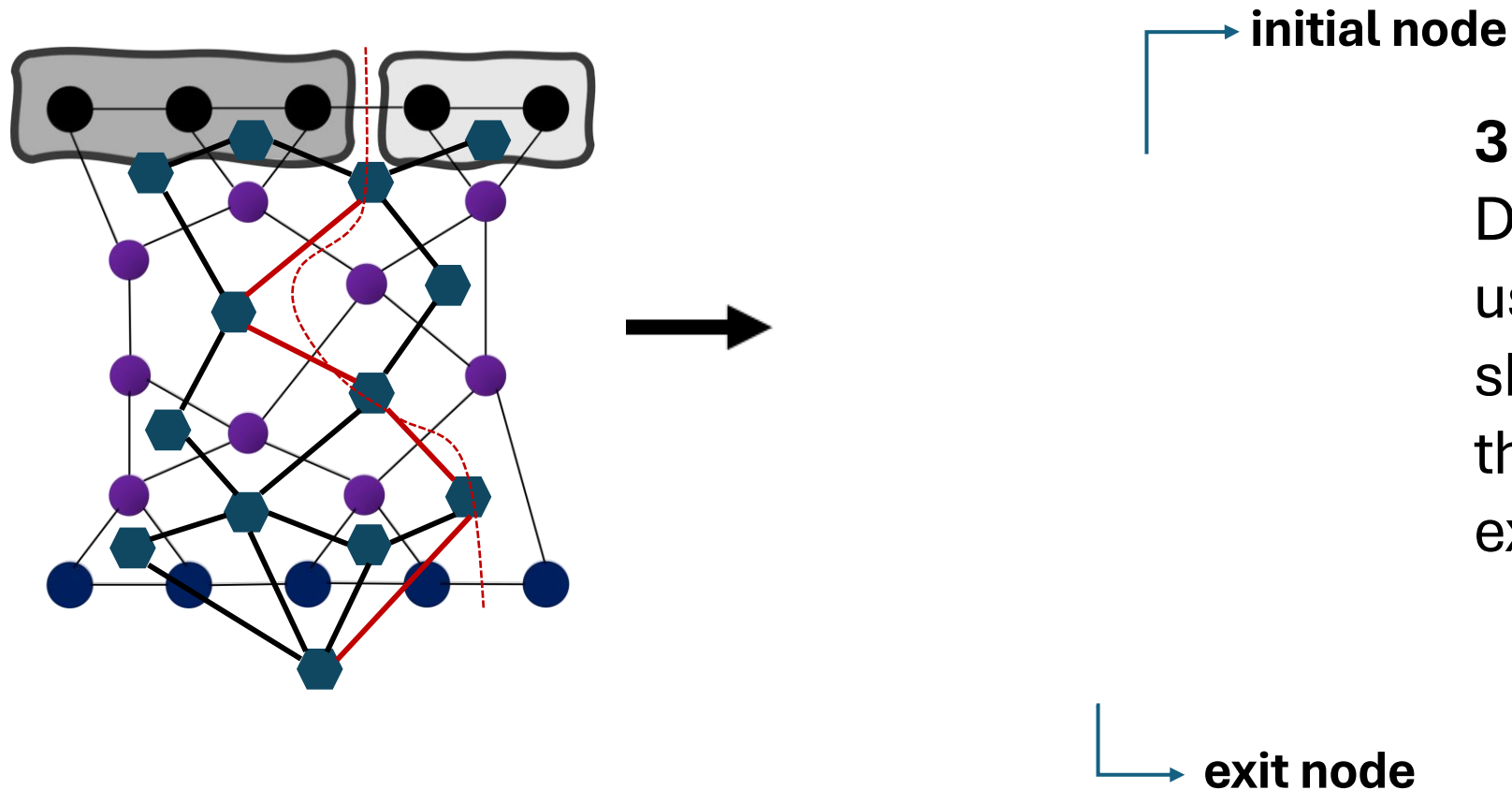
ALGORITHM FOR MINIMAL CUT

2. Another graph is constructed as a representation in cycle basis of the original.



ALGORITHM FOR MINIMAL CUT

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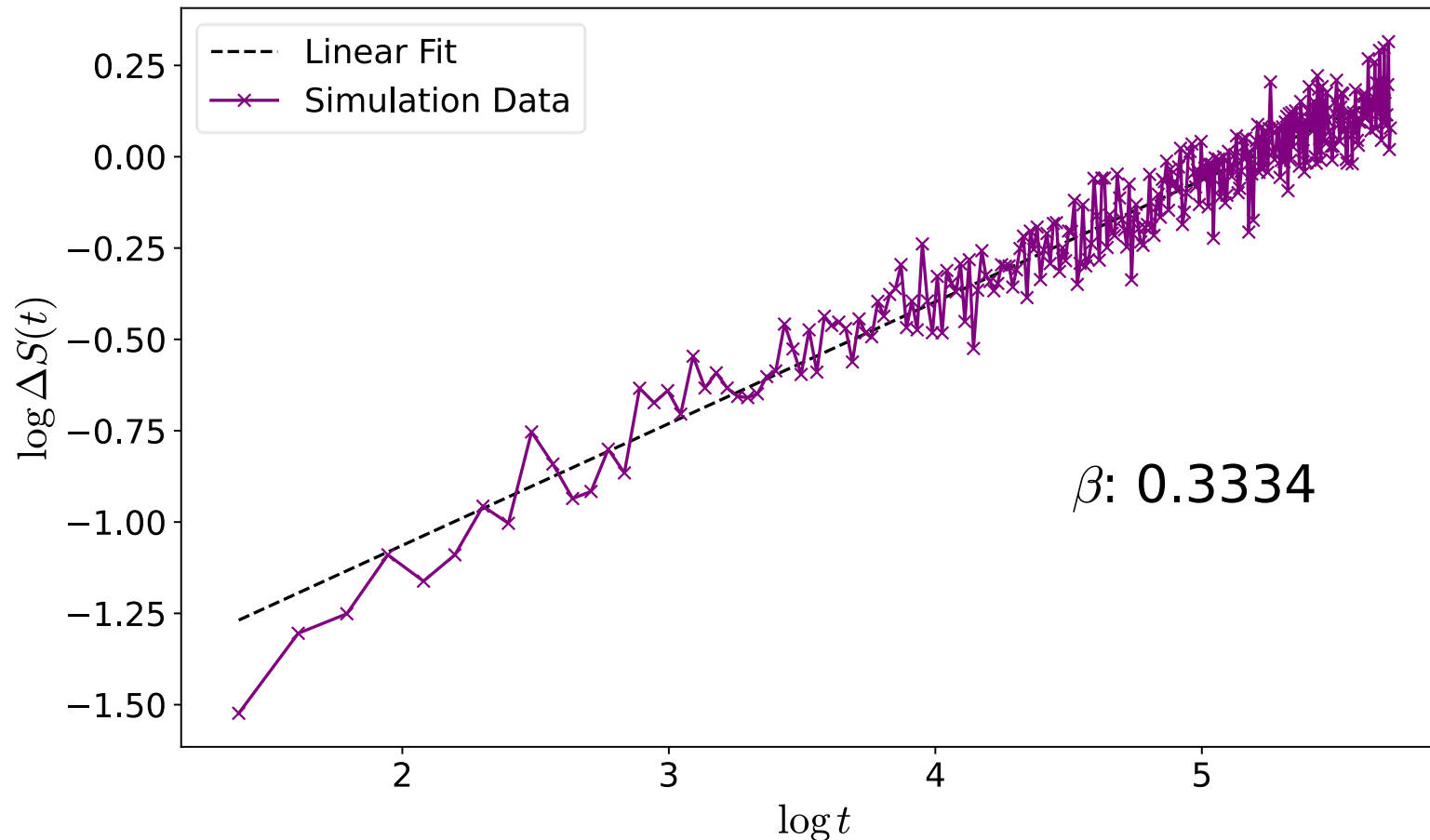


3. In this new graph, Dijkstra's algorithm is used to find the shortest path between the initial node and the exit node.

This is a deterministic algorithm, and the runtime is $O(V^2)$, where $V = \# \text{ qubits} + \# \text{ gates}$, i.e. number of nodes.

ALGORITHM FOR MINIMAL CUT

The KPZ coefficients can be found using this algorithm, which indicates that it works well.



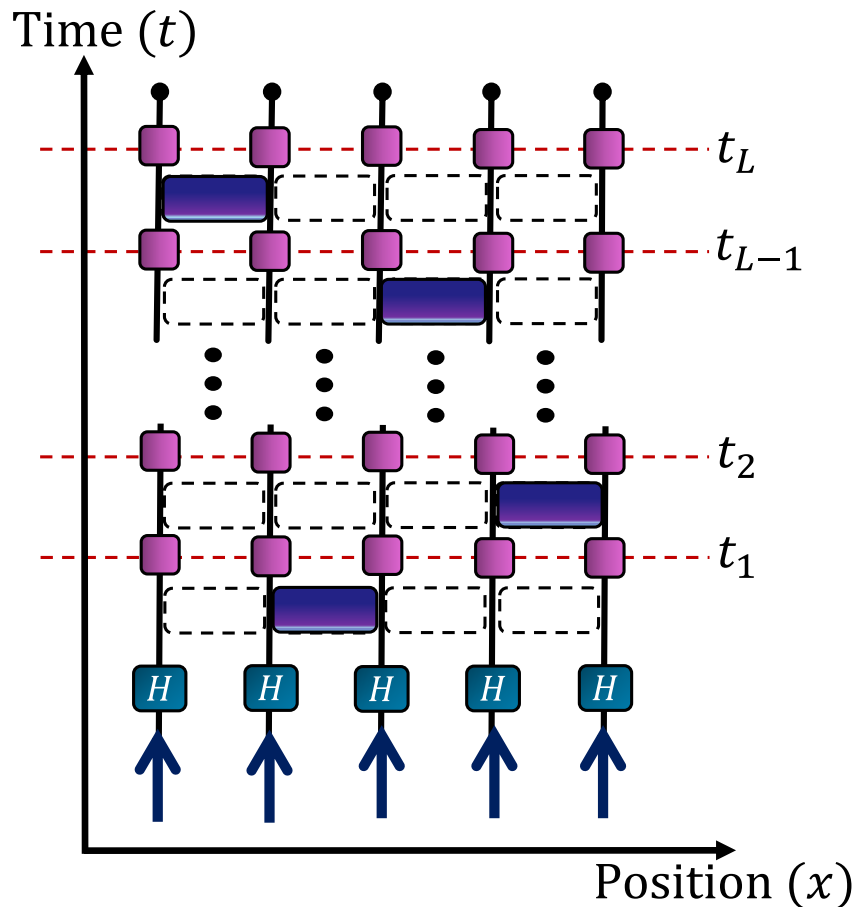
$$\Delta S(x, t) \sim t^\beta$$

A simulation is carried out for the growth of a classic graph with 50 qubits, until time $t = 300$.

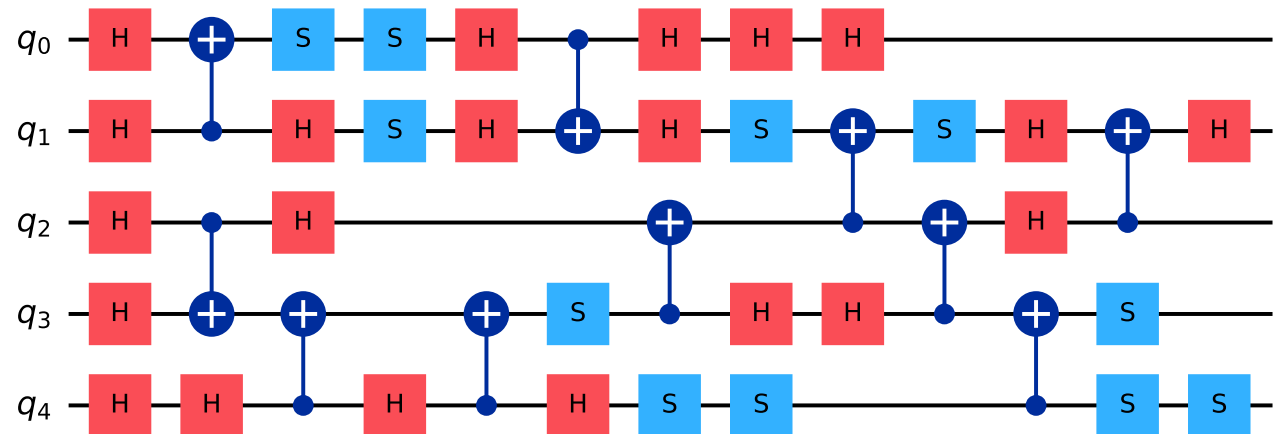
$$\beta \approx 1/3$$

SIMULATION

Clifford circuits, or “stabilizer circuits,” are a special class of quantum circuits that play an important role in quantum information theory.



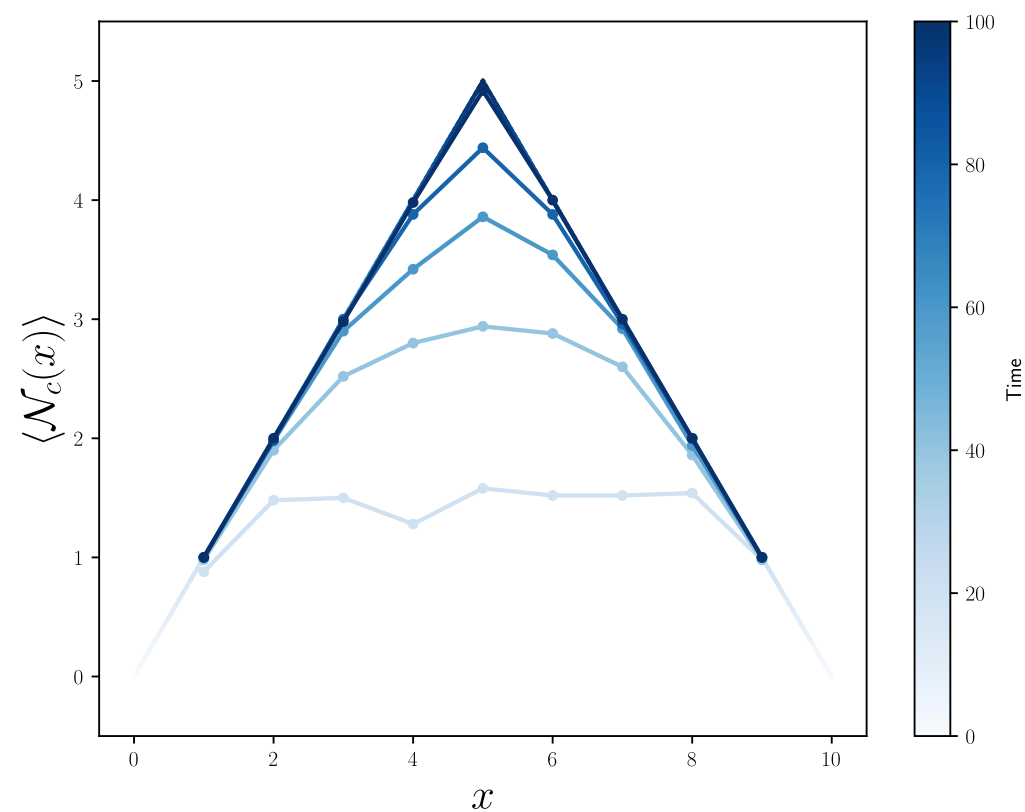
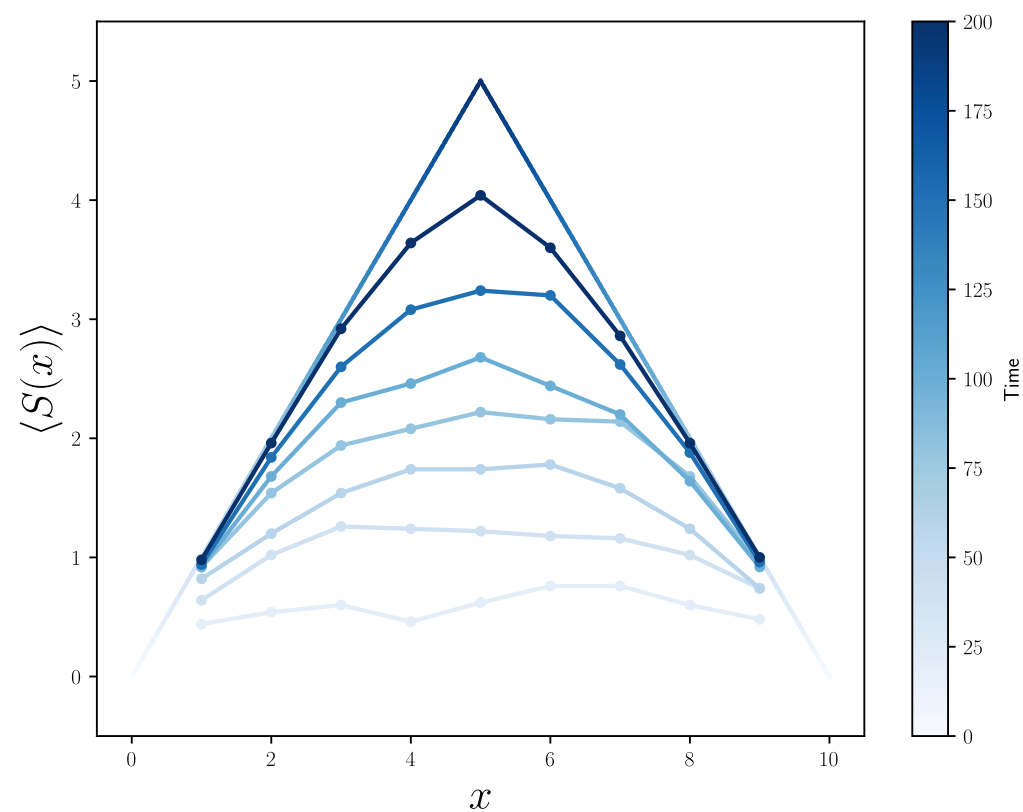
The Clifford group may be generated by a small set of local Clifford gates: the two-site **controlled NOT gates** and the **single-site Hadamard** and **phase gates**.



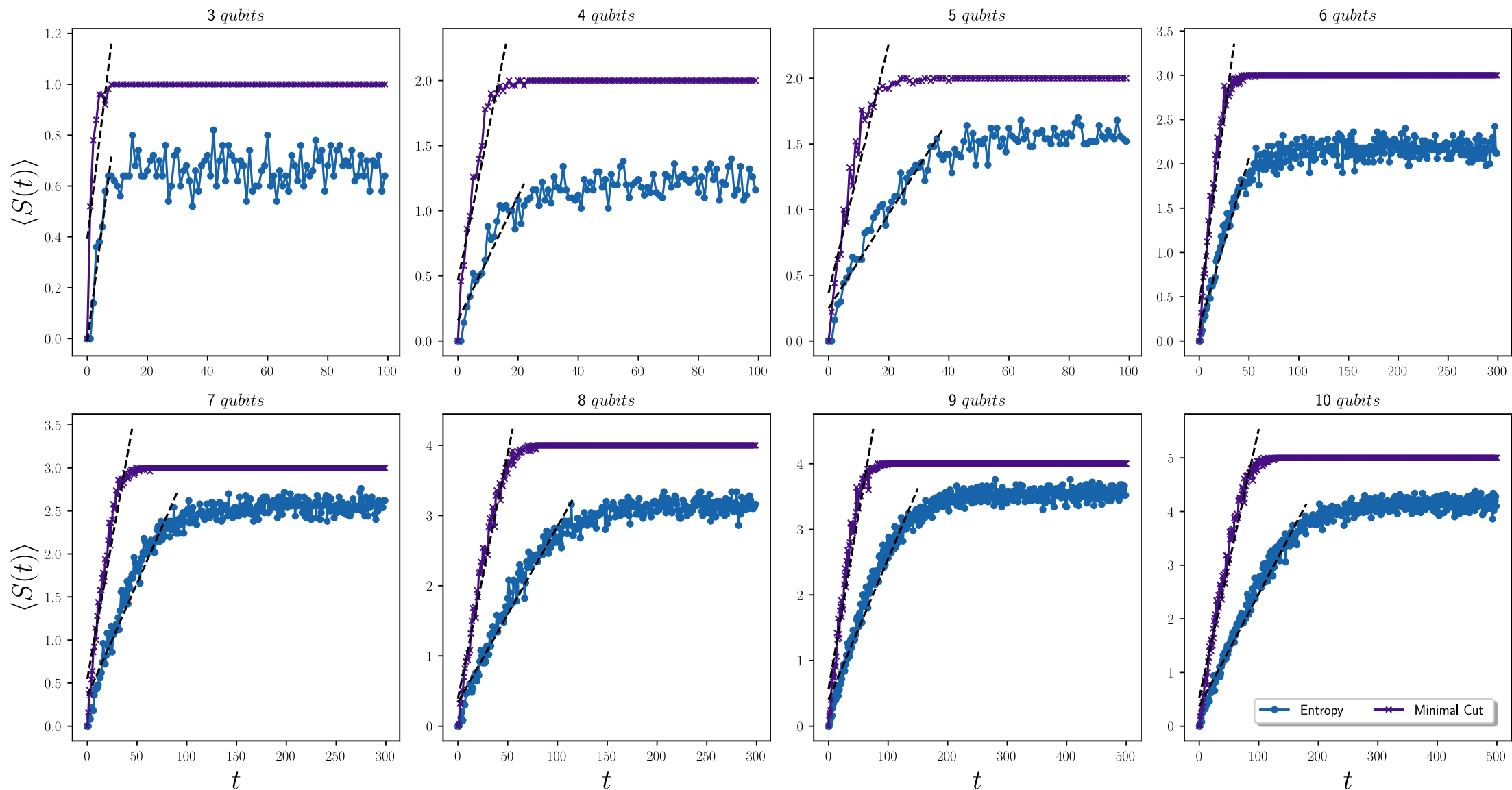
This is an example in Qiskit of a circuit with 5 qubits evolving until $t = 10$.

SIMULATION

The average entropy over a set of 50 random circuits over time versus position for a 10-spin system. The entropy is calculated for S_1 .

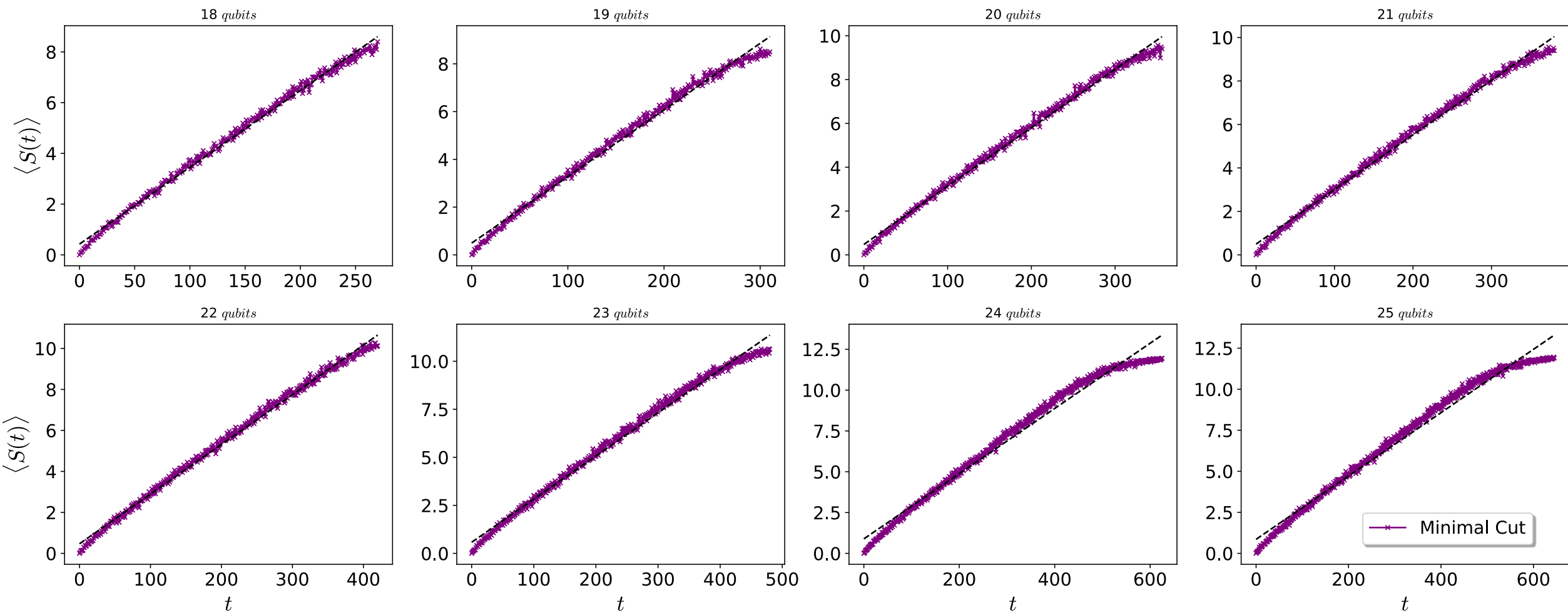


SIMULATION



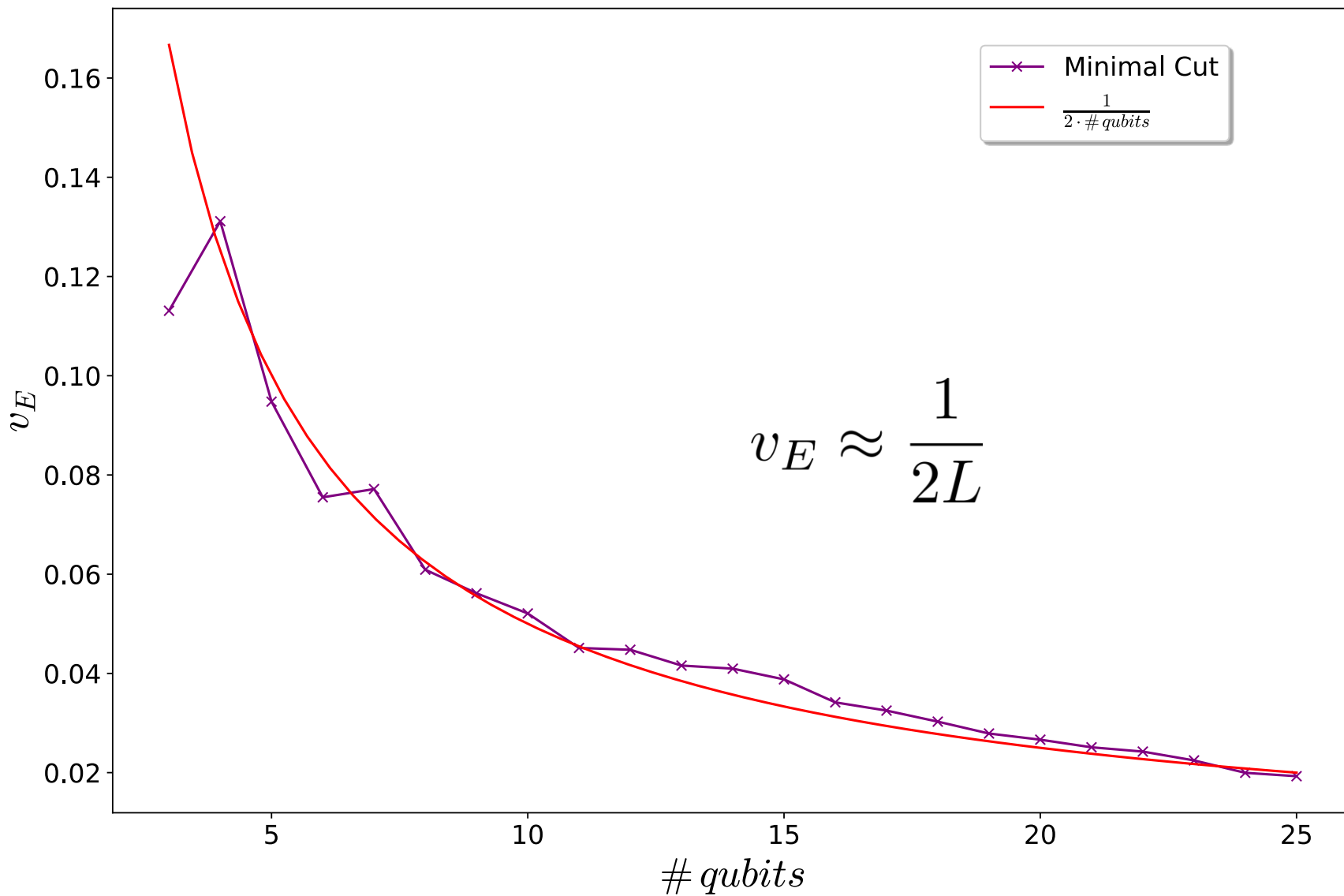
Average entropy dynamics at $x = L/2$, over a set of 100 random circuits.

SIMULATION



Average minimal cut dynamics at $x = L/2$, over a set of 100 random circuits.

SIMULATION



Remembering:

$$\langle \mathcal{N}_{\text{m.c.}}(t) \rangle = v_E t \sim \frac{t}{L}$$

$$t_{\text{saturation}} = \frac{L}{2v_E}$$

The numerical
results suggest



$$\langle \mathcal{N}_{\text{m.c.}}(t) \rangle \approx \frac{t}{2L}$$

$$t_{\text{saturation}} \approx L^2$$

CONCLUSIONS

- ❑ The asymptotic behavior of entropy is described by the minimal cut as,

$$\langle S(x, t) \rangle \approx \frac{t}{2L}.$$

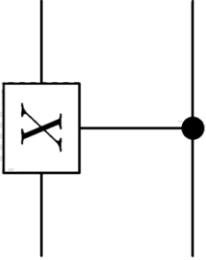
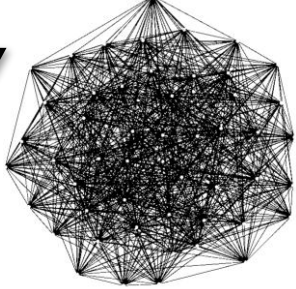
- ❑ The system reaches the maximally entangled state for L^2 , that is, when the density of qubits per gate is of the order of L .

- ❑ The *cut growth model* is a good model.



PERSPECTIVES

- ❑ Apply the model to other gates distribution, or specific architectures.
- ❑ Explore the model by doping the system with 3-qubit gates.

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